BSSE FINAL PROJECT
Software Design Specification

Investmate Nexus

Project Advisor

Zain Asghar

Presented by:
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Faculty of Information Technology & Computer Science

University of Central Punjab

University of Central Punjab

Software Design Specification

SDP Phase II

Investmate Nexus

Advisor: Zain Asghar

Team S25SE041

Member Name	MemberID	Primary Responsibility
Amber Abbas	L1S22BSSE0030	Frontend, backend
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Revision History

Name	Date	Reason For Changes	Version
Esha Afreen	18-5-25	Initial draft created for Phase 2 sections.	1.0
Amber Abbas	19-5-25	Created Activity Diagram, ERD, Decision Table, Collaboration Diagram, Architecture Diagram	1.1
Maniha	21-5-25	Created Sequence Diagram, Data Flow Diagram, Component Diagram, Event Trace Diagram, Class Diagram	1.2`
Esha Afreen	23-5-25	Created five screens AI Recommendations Messaging interface, Video call UI(Twilio), Payment form(Payoneer), Leaderboard.	1.3
Fiza Wajid	27-5-25	Created five screens Login/Register page, Dashboard Investor, Dashboard Entrepreneur, Submit Business Idea, Portfolio Tracker	1.4
Amber Abbas	2-6-25	Added technical Architecture	1.5
Maniha	3-6-25	Added Reference, revised project plan and other design details.	1.6
Fiza Wajid	3-6-25	Added overall description	1.7
Esha Afreen, Fiza Wajid, Amber Abbas, Maniha	17-6-25	Finalized Phase 2 Documentation	1.8

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

S25 Rejected Idea Defence Feedback + ... Download

PO Project Office<projects.fit@ucp.edu.pk>

To: L1S22BSSE0216 - ESHA AFREEN; L1S22BSSE0222 - MANIHA; L1S22BSSE0187 - FIZA WAJID; L1S22BSSE0030 - AMBER ABBAS BUTT Sat 5/10/2025 10:17 PM

Cc: Dr Adnan Ghafoor; Zain Asghar

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	**SE-16**
Project Title	**INVESTMATE NEXUS**

They are much clear about the concerns. Still a lot features, they need to work hard.

They have done good work.

Prepare your Proposal Document in the light of comments provided. Last date to submit the Proposal Document is **Wednesday, April 09, 2025** not later than **03:00 PM**. The proposal document should be prepared using the template available at http://smrms.ucp.edu.pk/All_Phases_Templates.html. Submit the hard copy to Project Office in F-304 (Main Building).

When submitting the hard copy of the proposal document, use the duly signed Group Allocation Yellow Page (available at http://smrms.ucp.edu.pk/Forms.html) as the first page of the submission document. Also get the proposal tape bind before submission.

Abstract

Investmate Nexus is a smart web platform developed to bridge the gap between investors and entrepreneurs, especially in underserved markets like Pakistan. The platform provides a secure, AI-driven ecosystem where entrepreneurs can present their business ideas while investors can discover, assess, and track potential investments. AI algorithms ensure accurate matchmaking, validate idea documents, and perform risk analysis to support informed decisions. Key features include role-based access, real-time messaging and video calls, investment portfolio dashboards, leaderboards for top ideas, and secure payment integrations. This document outlines the detailed design, architecture, diagrams, system workflows, and implementation plan for Phase II of the Investmate Nexus platform

1. Introduction

1.1 Product

Investmate Nexus is a smart, AI-powered matchmaking and investment collaboration platform designed to create real value for two main user roles: Entrepreneurs and Investors.

Features for Entrepreneurs:

- Upload and present business ideas with supporting documents (e.g., licenses, feasibility reports).
- Use AI tools for idea validation and scoring.
- Gain exposure through leaderboards and badges.

Features for Investors:

- Receive AI-powered recommendations tailored to their interests.
- Manage their investments via an intuitive portfolio dashboard.
- Assess idea risk and market value before making decisions.
- Communicate with entrepreneurs directly through integrated messaging and video calling.

1.2 Background

There is a significant disconnect in developing markets like Pakistan between aspiring entrepreneurs and verified investors. Current platforms offer either limited exposure or basic funding methods without the necessary intelligence or collaboration features.

Global Tools:

- AngelList: Good for startup listings, lacks AI personalization.
- Kickstarter: Focuses on donation/crowdfunding instead of long-term partnerships.
- SeedInvest: Better for funding, but minimal risk analysis or direct interactions

Local Tools:

- Pak Angels & i2i: Provide funding opportunities, but don't automate investor matching or idea validation.

Investmate Nexus aims to offer a comprehensive alternative, combining intelligent recommendations, validation systems, investment tracking, and integrated communication under one roof.

1.3 Objective(s)/Aim(s)/Target(s)

The core objectives of Investmate Nexus are:

- Enable Role-Based Access:
  - Users can register as Entrepreneurs or Investors.
  - Each role sees only the features and tools that are relevant to their needs.
- Intelligent AI Matchmaking:
  - Recommend investment ideas based on investor history, preferences, and platform interactions.
- Idea Validation and Scoring System:
  - Entrepreneurs upload documents that are scanned and scored based on criteria like scalability, competition, and business model.
- Risk Analysis for Investments:
  - Predictive tools that assess market conditions, idea strength, and projected ROI.

- Interactive Communication Suite:
  - Investors and entrepreneurs connect using chat, video conferencing, and alerts.
- Investment Tracking and Reports:
  - Dashboards for investors showing investment health, trends, and analytics.
- Leaderboards and Engagement Rewards:
  - Users get featured for active contributions.
- Secure Payments and Transparency:
  - Integration with trusted financial services to make and track transactions.

1.4 Scope

The scope includes:

- Authentication & role assignment
- Profile and dashboard customization
- Idea submission and AI validation
- Investment recommendations engine
- Risk prediction models
- Communication module (chat & video)
- Payment and subscription module
- Leaderboards and recognition

1.5 Business Goals

The business goals are:

- Foster innovation and entrepreneurship
- Build investor trust through transparency and risk analysis
- Promote startup culture in Pakistan
- Encourage collaboration and learning through leaderboards
- Generate revenue via freemium/premium models

1.6 Document Conventions

- Use of bold text for headings.
- Technical terms are capitalized.
- Use of consistent numbering and bullet points for clarity.

1.7 Miscellaneous

This SDS is designed to ensure clarity of system behavior, architecture, and design. Diagrams are created in StarUML and Figma.

2. Overall Description

2.1 Product Features

Investmate Nexus offers the following high-level features:

- User Role Selection & Authentication
  Entrepreneurs and Investors register separately and access features relevant to their roles through secure login.
- AI-Based Recommendations
  Investors receive business idea suggestions based on interest areas, history, and market trends using intelligent algorithms.
- Business Idea Submission
  Entrepreneurs upload pitch decks, licenses, business plans, and videos for investor review.
- Idea Scoring & Risk Analysis
  Ideas are validated and scored automatically by the system based on financial soundness, market competition, and scalability.
- Real-Time Communication Tools
  In-app chat and video calling (via Twilio API) foster interactive discussions and negotiations.
- Investment Portfolio Management
  Investors can view and monitor their investments through interactive dashboards and performance analytics.
- Gamification
  Leaderboards highlight top contributors and most active users. Badges reward milestones like successful deals and verified documents.
- Secure Payment Integration
  Integration with Payoneer ensures safe and transparent investment transactions.

2.2 Functional Description

Investmate Nexus is a web-based platform designed to facilitate smart, AI-driven matchmaking between entrepreneurs and investors, especially in developing markets like Pakistan. The system enables users to register under role-based access, submit or review business ideas, communicate in real-time, and securely manage investment transactions.

Core Functionalities:

- Role-Based User Access: Users register as either Entrepreneurs or Investors with different feature access.
- Entrepreneur Features:
  - Submit business ideas with documentation (pitch deck, license, video).
  - Use AI tools for idea validation and scoring.
  - View feedback and leaderboard rankings.
- Investor Features:
  - Receive AI-powered idea recommendations.
  - Analyze risk and validate submissions.
  - Manage investment portfolio.
  - Communicate with entrepreneurs via chat/video.
  - Make secure payments.

- AI Layer:
  - Matches investor preferences with entrepreneur ideas.
  - Performs risk analysis and scoring.
- Communication Tools:
  - Integrated messaging and video calling (via Twilio/WebRTC).
  - Payment Integration:
  - Secure payment processing using services like Payoneer.

2.3 User Classes and Characteristics

Investmate Nexus caters to four main user classes. Each class interacts with the platform in distinct ways, and the system is primarily optimized for the favored user groups, while still offering limited support for others.

1. Entrepreneur(Favored User Class)

- Role: Business idea proposer and project submitter.
- Characteristics:
  - May not have deep technical knowledge.
  - Familiar with documents like feasibility studies and licenses.
  - Highly motivated to receive funding and recognition.
- Needs:
  - Business idea submission with document upload.
  - AI-based idea validation and scoring.
  - Access to messaging/video chat tools for investor communication.
  - Visibility through leaderboards and promotions.

2. Investor(Favored User Class)

- Role: Evaluates and funds business ideas.
- Characteristics:
  - Experience in business analysis and funding.
  - Risk-averse and analytical.
  - Prefers intelligent suggestions and data-driven decisions.
- Needs:
  - Personalized recommendations through AI.
  - Risk evaluation tools and return-on-investment insights.
  - Access to secure chat, call, and portfolio management tools.
  - Investment tracking and reporting features.

3. Admin(Secondary User Class, Less Important to Satisfy)

- Role: System manager and content moderator.
- Characteristics:
  - Technically proficient with backend access.
- Needs:
  - Manage and monitor users and content.
  - Review submitted business ideas.
  - Oversee system logs and reports.
  - Although essential for platform governance, Admins are not the primary focus of UX optimization. Their interfaces are functional rather than user-friendly.

4. Guest (Unregistered Visitor)(Less Important to Satisfy)

- Role: Visitor who has not registered yet.
- Characteristics:
  - May be a potential future user.
  - Limited system access.
- Needs:
  - View basic public content.
  - Option to register or explore platform features.
  - Learn about benefits of joining the platform.
  - Guests are not prioritized in system design, as they are expected to convert to registered users to gain full functionality.

2.4 Design and Implementation Constraints

- Technology Stack:
  - Frontend: HTML, CSS, Bootstrap, JavaScript
  - Backend: Java (Spring Boot)
  - AI: Python (Flask/FastAPI services)
  - Database: MySQL (Relational)
  - APIs: Twilio (Communication), Payoneer (Payments)
- Hosting Limitations:
  - Hosted on a shared cloud environment (initially); may scale later.
- Security Constraints:
  - Role-based authorization required for all endpoints.
  - JWT-based token authentication.
  - Encrypted payment and message data.
- Document Uploads:
  - File size limited to 10MB per document.
  - Only specific formats allowed: PDF, DOCX, PNG, JPG.

2.5 Assumptions and Dependencies

- AI Model Accuracy:
  Assumes the AI recommendation system and risk analysis models are trained on diverse and accurate data. Future retraining may be needed.
- Third-Party Service Availability:
  Relies on consistent availability of Twilio, Payoneer, and cloud services. Downtime may affect features.
- User Honesty in Data Submission:
  Idea scoring and recommendation accuracy assume users provide valid and complete data.
- Browser Compatibility:
  Optimized for modern browsers (Chrome, Edge, Firefox). Older versions may face UI issues.
- Dependencies:
  - Python libraries for AI: scikit-learn, pandas, transformers
  - Java libraries for Spring Boot: Spring Security, JPA, Thymeleaf

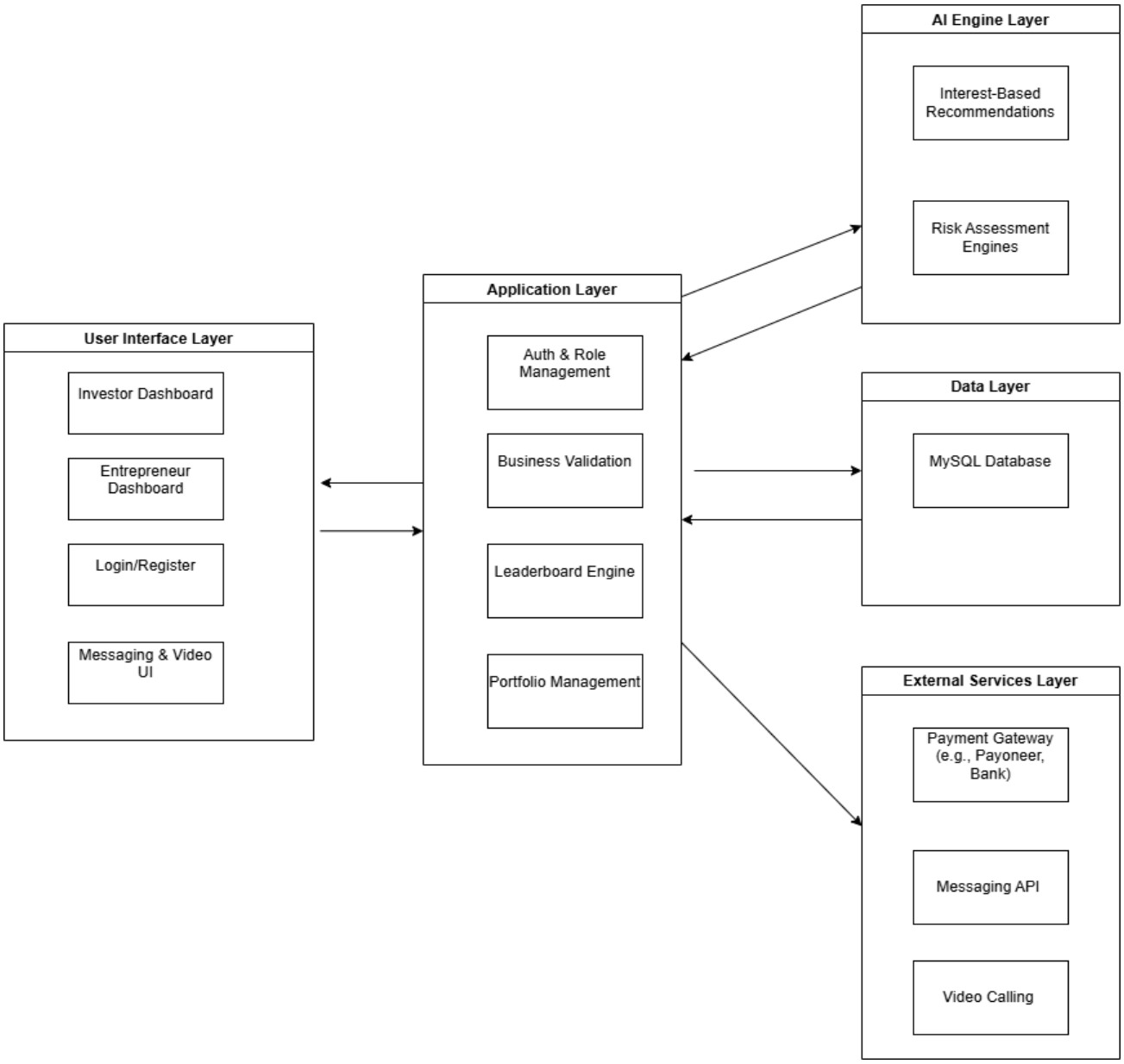
3. Technical Architecture

- The system is a custom-built web application designed to connect investors and entrepreneurs.
- It supports both batch and online processing to handle user registrations, recommendations, and secure transactions.
- Includes transaction processing (investment, subscriptions) and analytical reporting (portfolio insights, leaderboards rankings).
- Major application components include:
  - AI Recommendation Engine (Python)
  - Role-Based Login System
  - Investor Portfolio Management Module
  - Business Idea Validation Engine
  - Risk Assessment Tool
  - Secure Payment Gateway Integration
  - Communication Module (Messaging & Video Conferencing)
- Data collected and managed includes user profiles, business ideas, investment portfolios, messages, payment logs, etc.
- Application follows a layered architecture with clear separation between frontend (HTML, Bootstrap, JS), backend (Java), and AI modules (Python).
- Hosted on an Internet-accessible platform with cloud-based deployment.
- Uses MySQL as the database platform.
- End-user interface is web-based (browser-based UI).
- Network architecture supports access overWAN/Internet.
- Hosting is anticipated on a cloud service like AWS, Heroku, or university-provided servers.

3.1 Application and Data Architecture

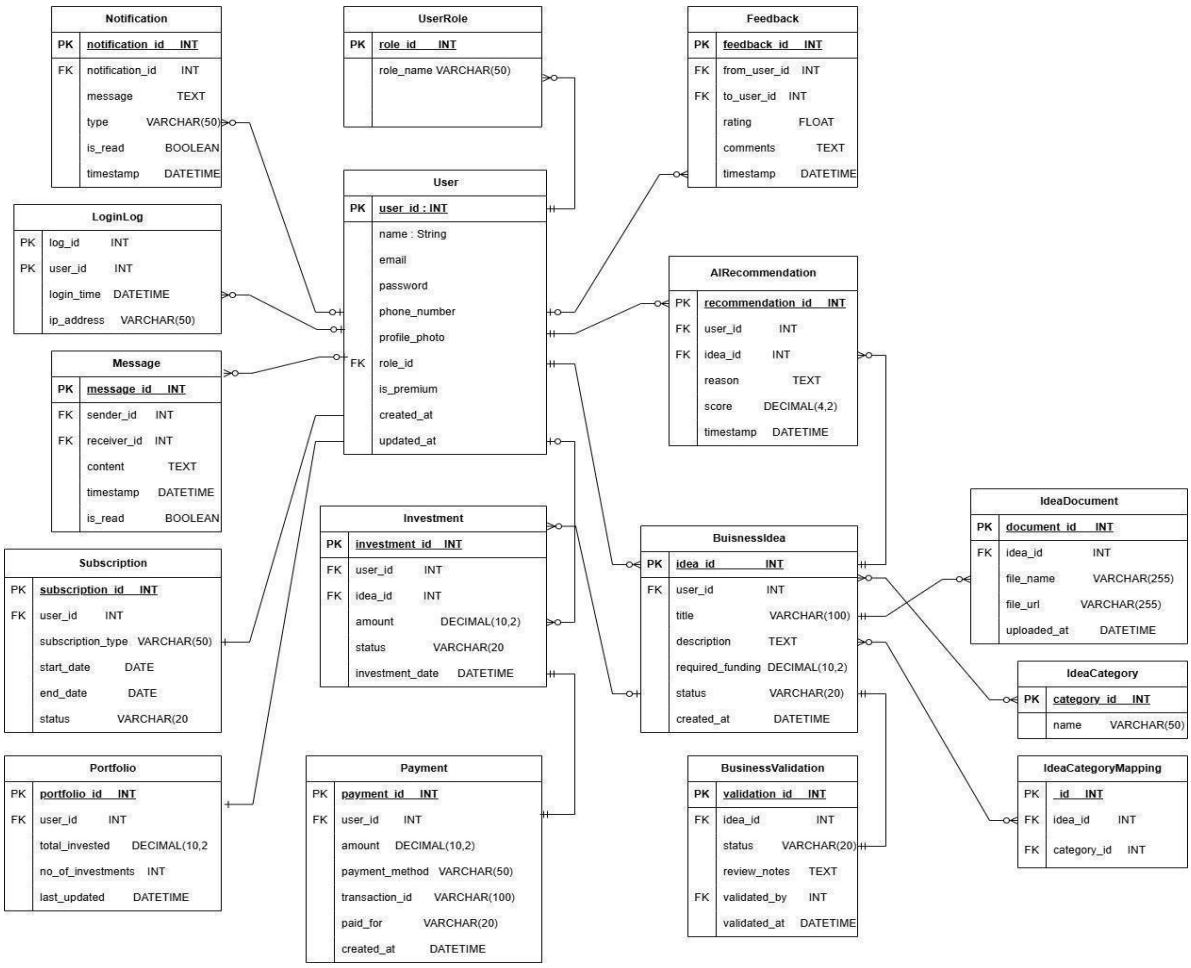
3.1.1. Architecture Diagram

This high-level architecture diagram illustrates the layered structure of the Investmate Nexus platform. It outlines the interaction between the user interface, application logic, AI engines, data storage, and external services such as payments and messaging.



3.1.2. ER Diagram

This Entity-Relationship Diagram (ERD) illustrates the logical structure of the database for Phase 2 of the FYP, depicting entities such as User, Business Idea, Investment, and Payment, along with their attributes and the relationships between them. It provides a blueprint for data organization and integrity within the system.



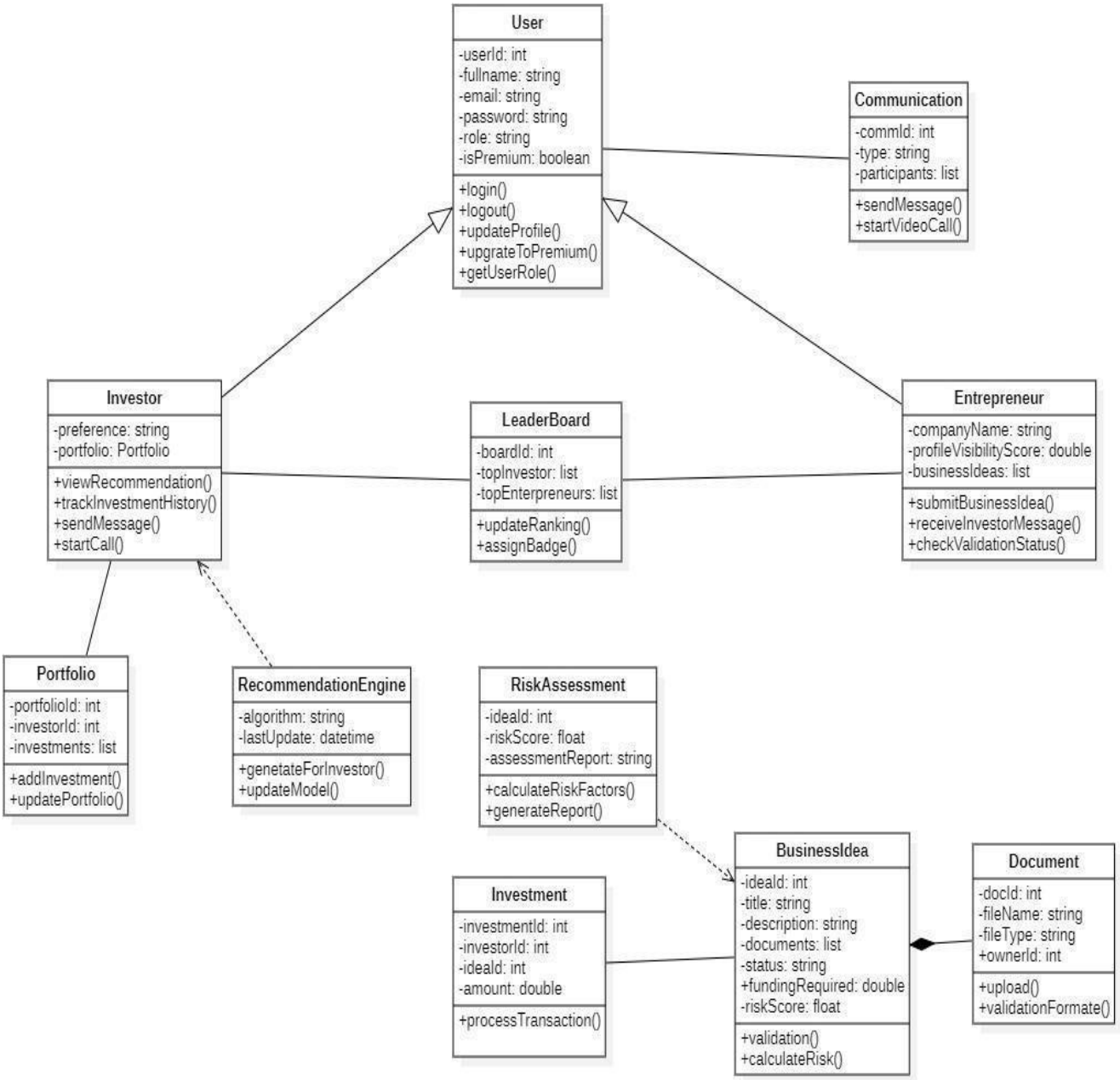
3.1.3. Decision Table

This decision table systematically outlines the system's behavior based on various user roles, profile completion, idea submission status, AI recommendations, risk assessments, and investor selections. It maps these conditions to specific actions, ensuring clear and consistent responses within the application's workflow.

Rule	User Role	Profile Complete	Idea Submitted	AI Recommends Idea	Risk Acceptable	Investor Selected Idea	Action to Perform
1	Investor	Yes	-	Yes	Yes	Yes	Allow investment process & open chat/payment
2	Investor	Yes	-	Yes	No	Yes	Show risk warning, ask to select new idea
3	Investor	Yes	-	No	-	-	Display 'No matches found', suggest new preferences
4	Investor	No	-	-	-	-	Prompt to complete investor profile
5	Entrepreneur	Yes	Yes	-	-	-	Send idea to AI for validation + wait for investor
6	Entrepreneur	Yes	No	-	-	-	Prompt to submit business idea
7	Entrepreneur	No	-	-	-	-	Prompt to complete entrepreneur profile
8	Entrepreneur	Yes	Yes	-	-	Idea Selected	Notify entrepreneur of investor interest
9	Investor	Yes	-	Yes	Yes	No	Wait for user to select idea
10	Entrepreneur	Yes	Yes	-	-	Not Selected Yet	Show 'Waiting for investor' status

3.1.4. Class Diagram

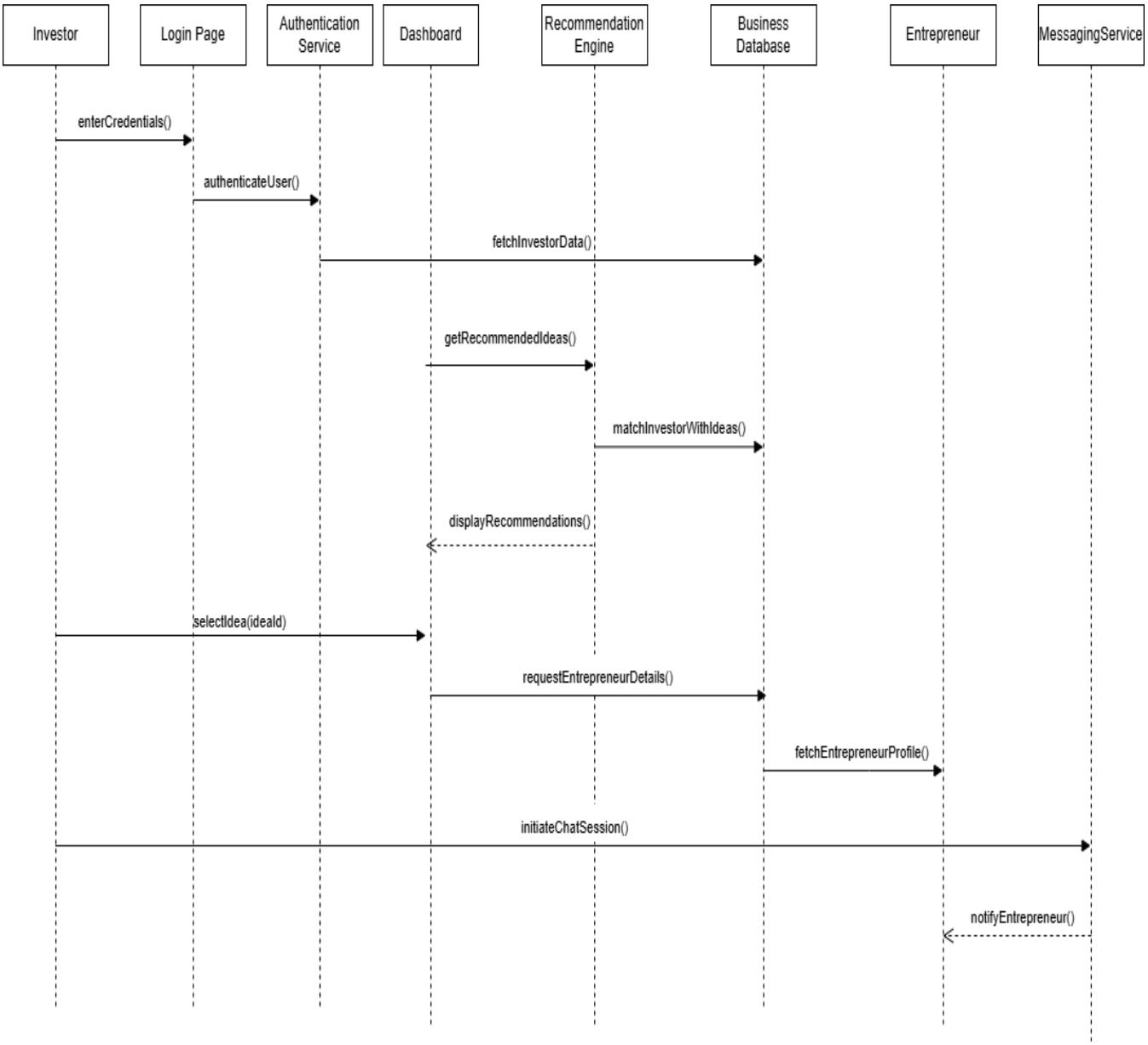
The class diagram defines system structure using classes such as User, Investor, Entrepreneur, BusinessIdea, and Portfolio.It shows their attributes, methods, and relationships like inheritance, composition, and associations.



3.2 Component Interactions and Collaborations

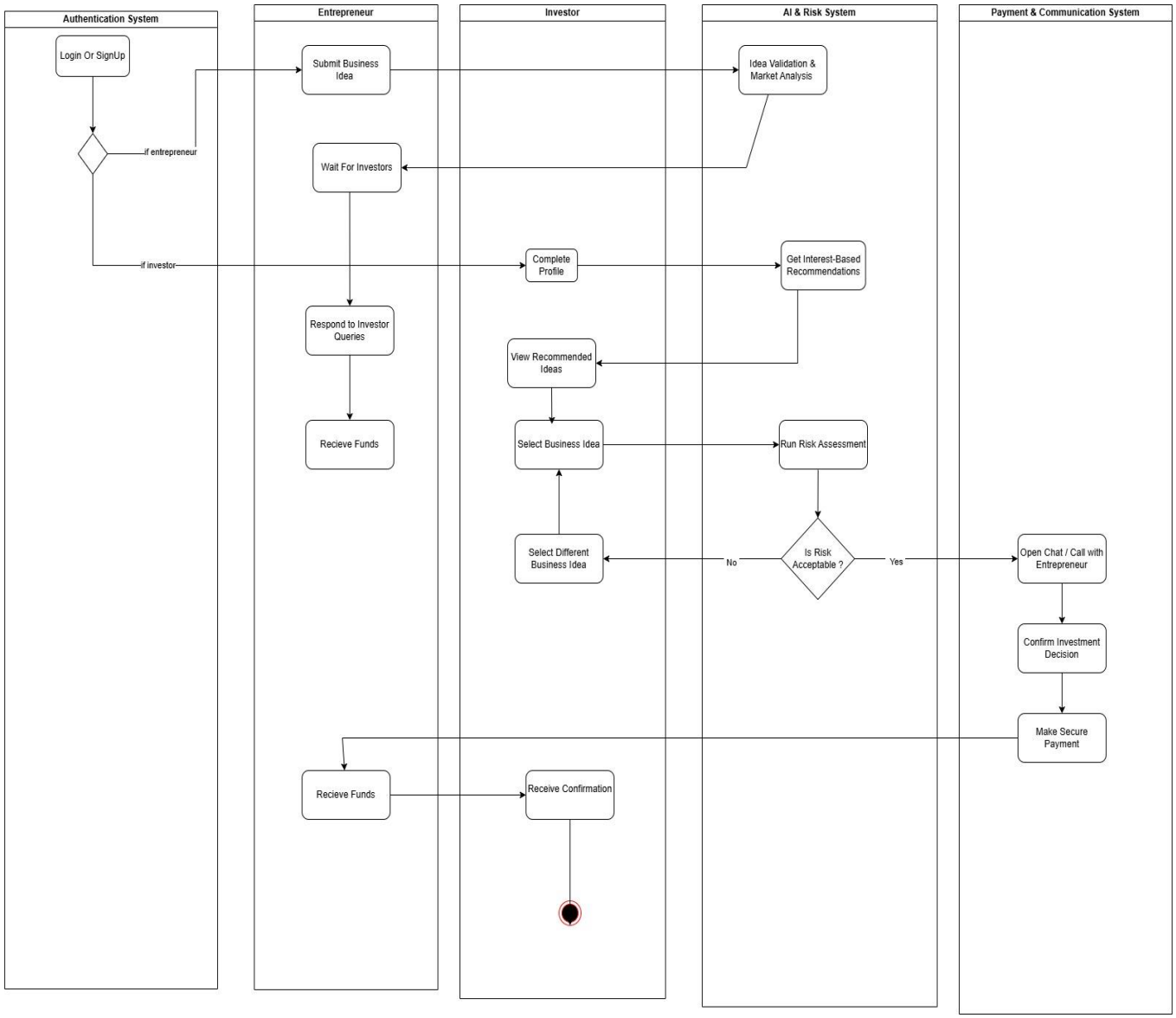
3.2.1. Collaboration diagram

This collaboration diagram demonstrates the step-by-step interaction between system components during the investor's journey. It focuses on the authentication process, recommendation flow, business idea selection, and communication initiation with entrepreneurs.



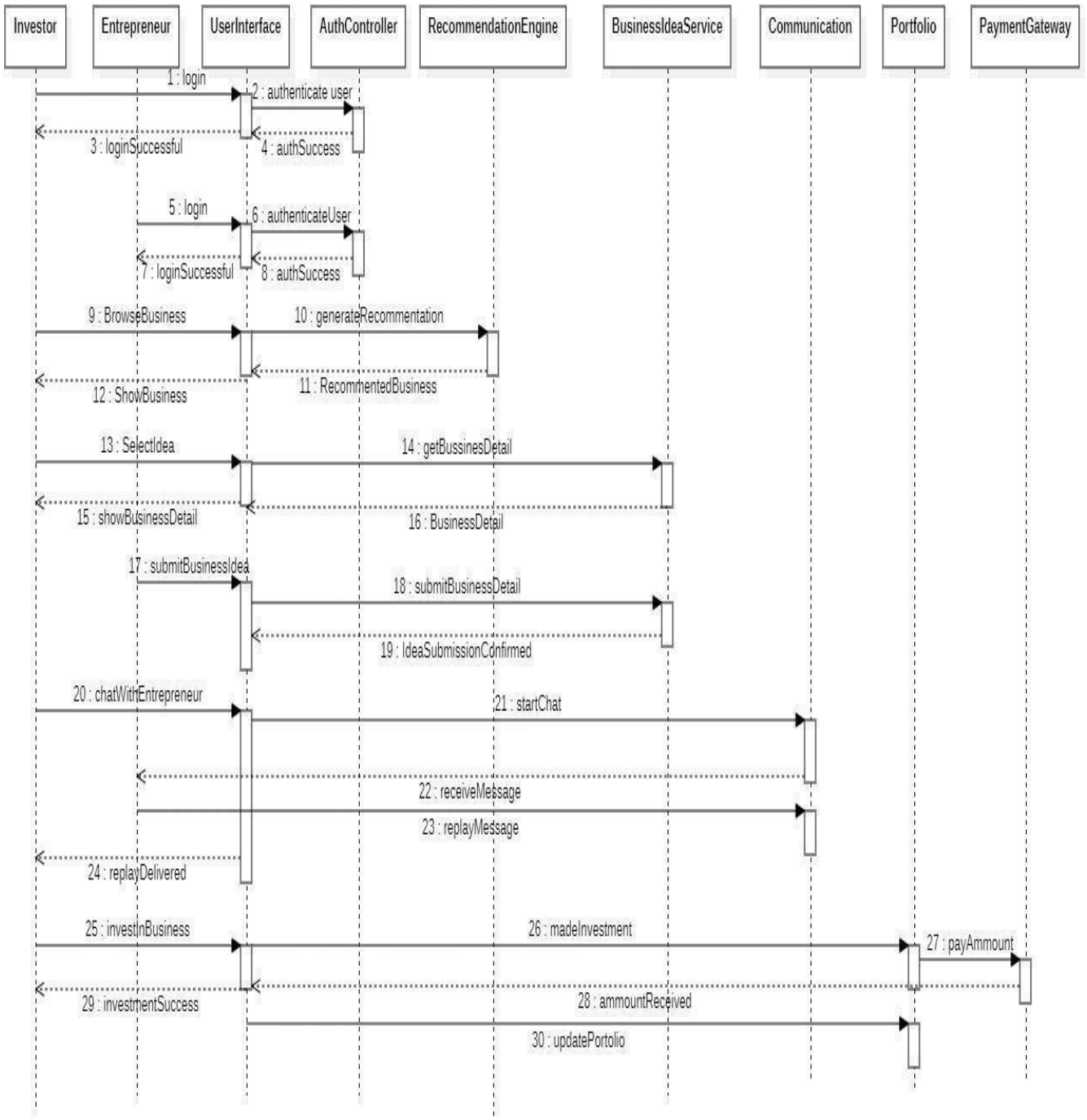
3.2.2. Activity Diagram

The activity diagram maps out the complete workflow from user login to final fund disbursement. It clearly separates the responsibilities across systems—Authentication, Investor, Entrepreneur, AI & Risk Analysis, and Payment/Communication—to visualize decision-making and process flow.



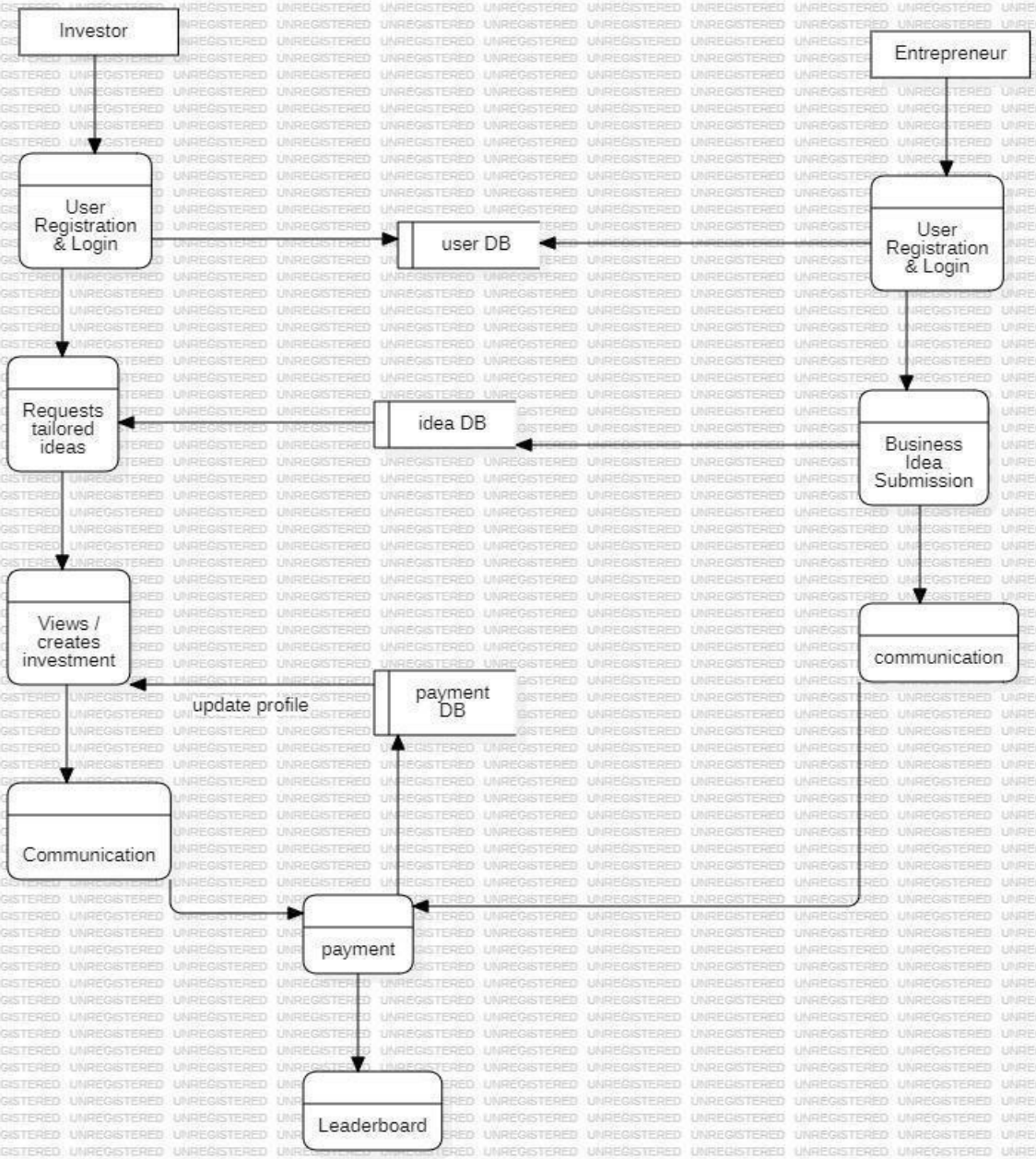
3.2.3. Sequence Diagram

The sequence diagram describes step-by-step message exchanges during the investment workflow. It includes login, recommendation retrieval, viewing business ideas, chatting, and completing payment.



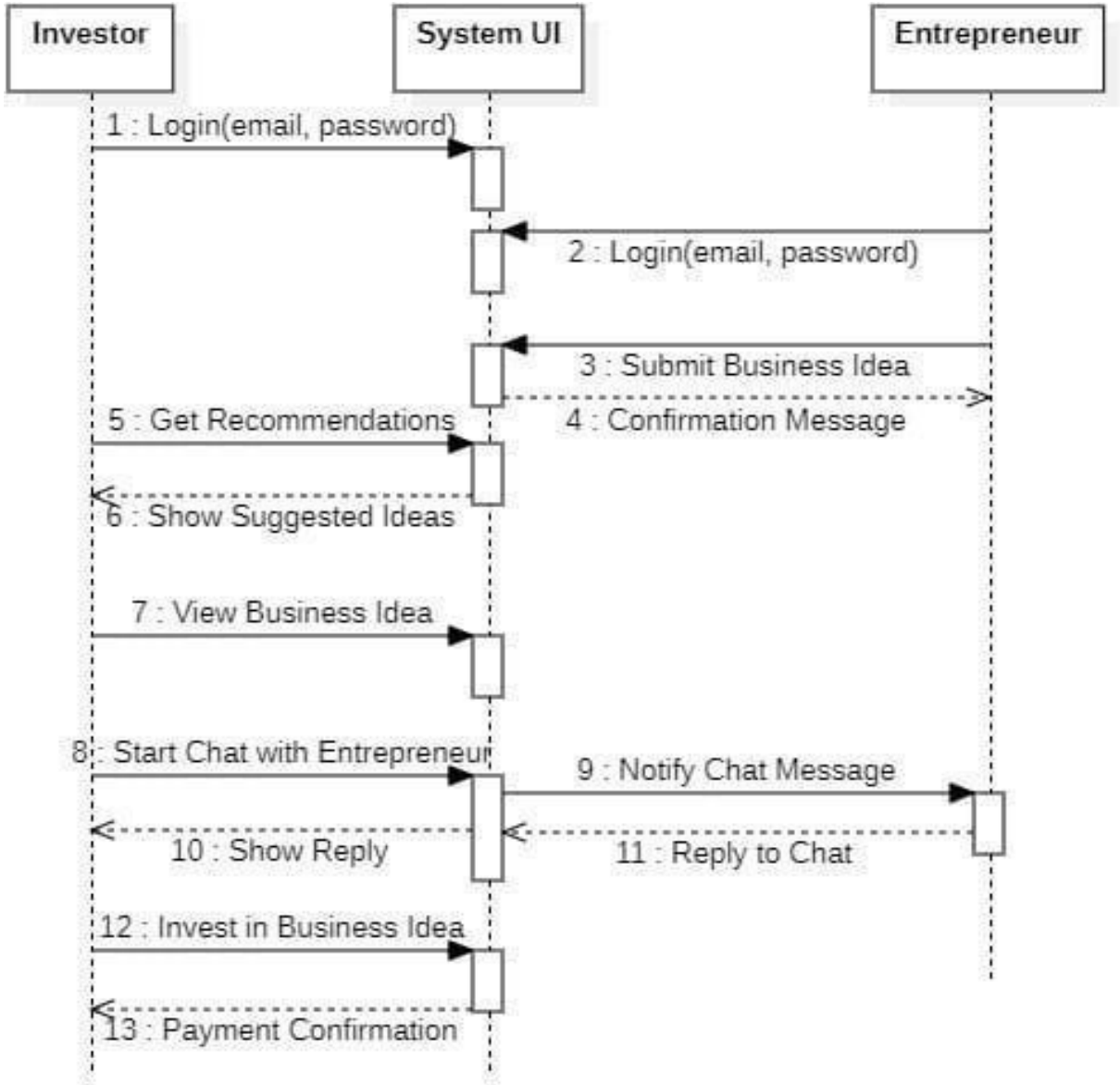
3.2.4. DFD Diagram

The DFD Level 1 breaks the system into processes like login, idea submission, investment, and chat. It shows how data flows between processes, external entities (users), and internal data stores.



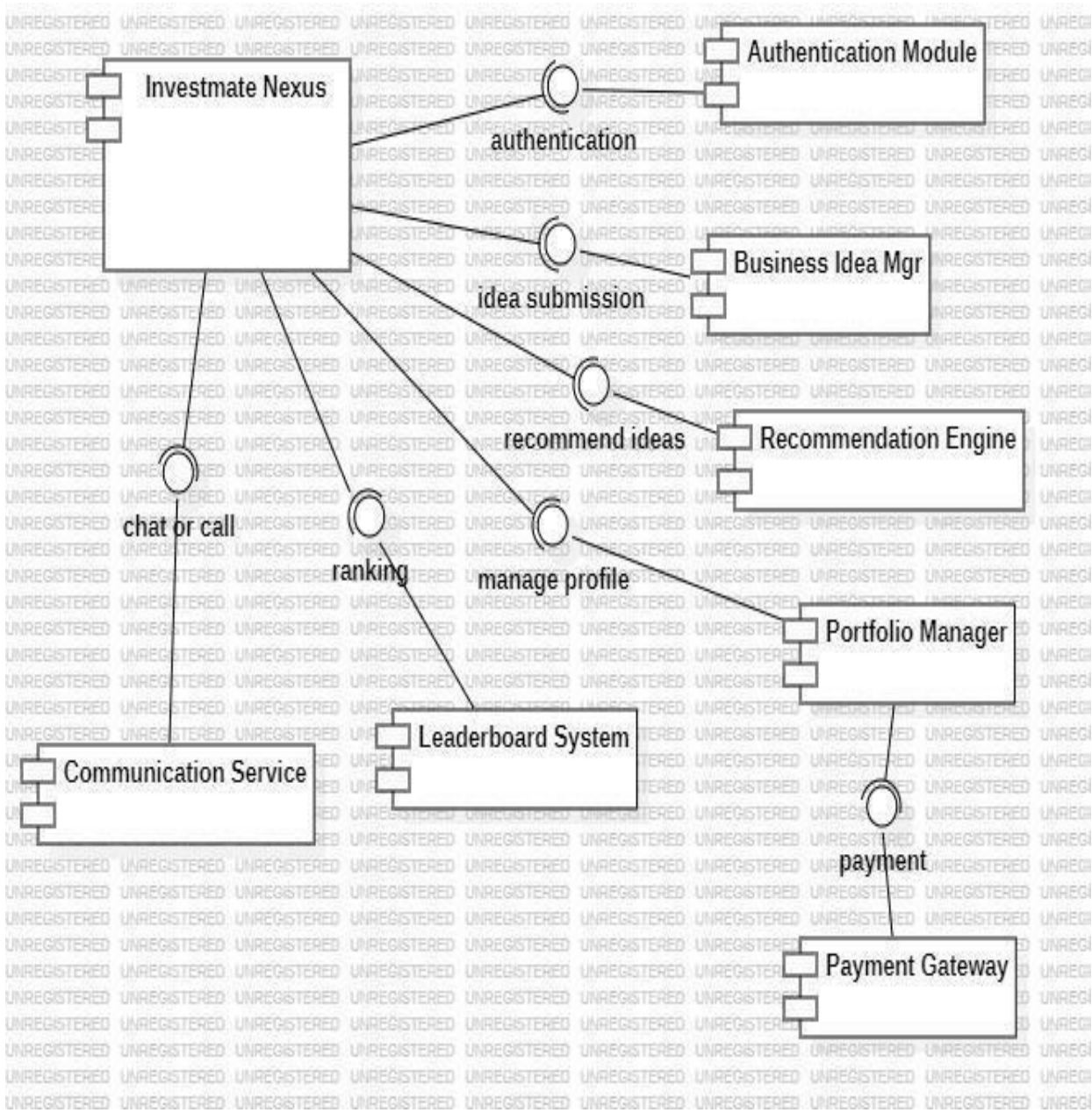
3.2.5. Event Trace Diagram

This diagram shows the interaction between Investor, Entrepreneur, and System UI over time.
It highlights the main user-level events like login, idea submission, recommendation, chat, and investment.



3.2.6. Component Diagram

The component diagram illustrates the modular structure of the system, separating frontend, services, and integrations.It shows how components like Authentication, Portfolio Manager, and Payment Gateway interact through interfaces.



3.3 Design Reuse and Design Patterns

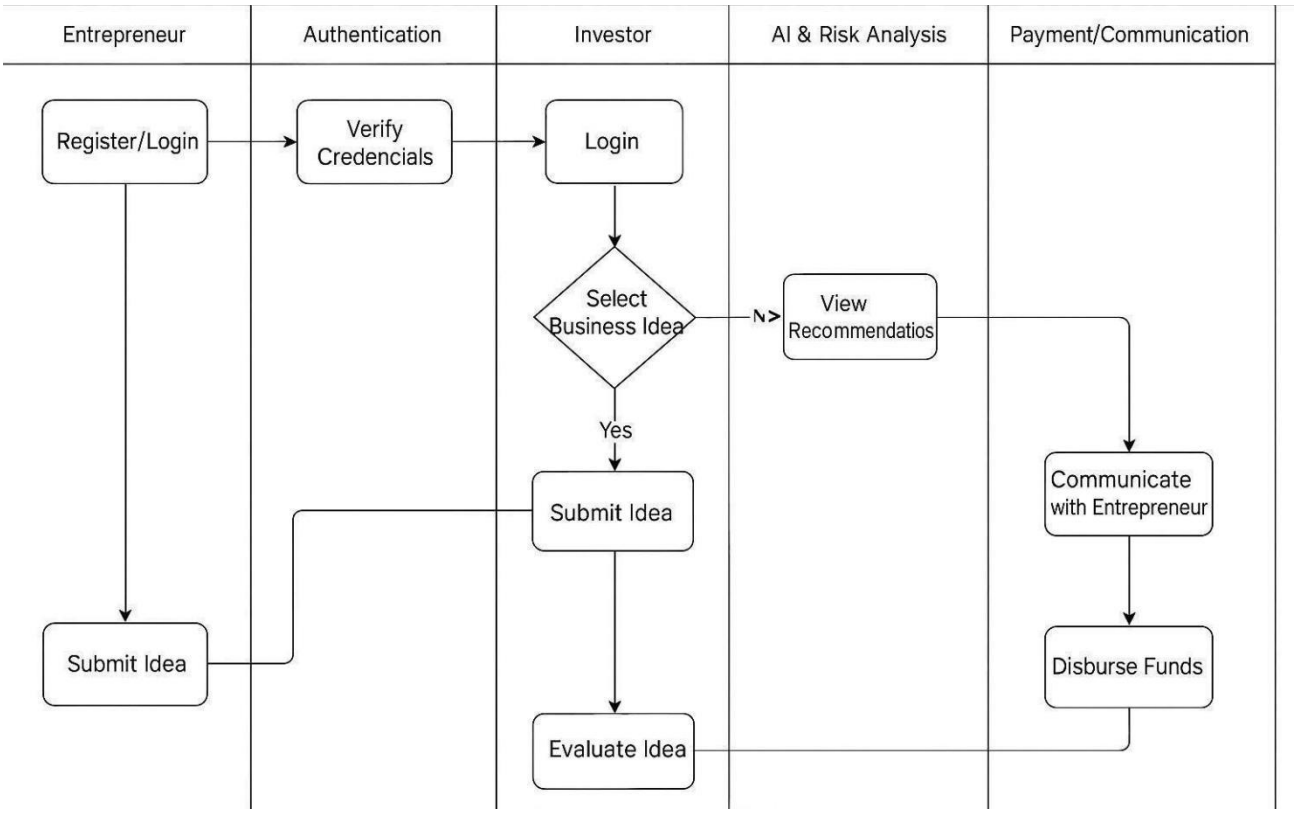
- Applied Patterns:
  - Singleton Pattern for portfolio manager
  - MVC Architecture (Model-View-Controller)
  - Observer Pattern for leaderboard updates and notifications
  - Factory Pattern for user role instantiation
- Reuse:
  - Bootstrap templates reused for frontend UI
  - Open-source Java packages for RESTful API handling and authentication

3.4 Technology Architecture

- Platform: Web-based platform accessible via browser
- Hosting: Cloud hosting (e.g., AWS, Heroku), university server (if internal)
- Connectivity: Requires stable Internet connection; accessible over secure HTTPS
- Operating Modes: Online mode with real-time communication and data syncing
- Tools & Technologies:
  - Frontend: HTML, Bootstrap, JS
  - Backend: Java (Servlets/APIs)
  - AI Layer: Python (for recommendations, validation, risk)
  - Database: MySQL

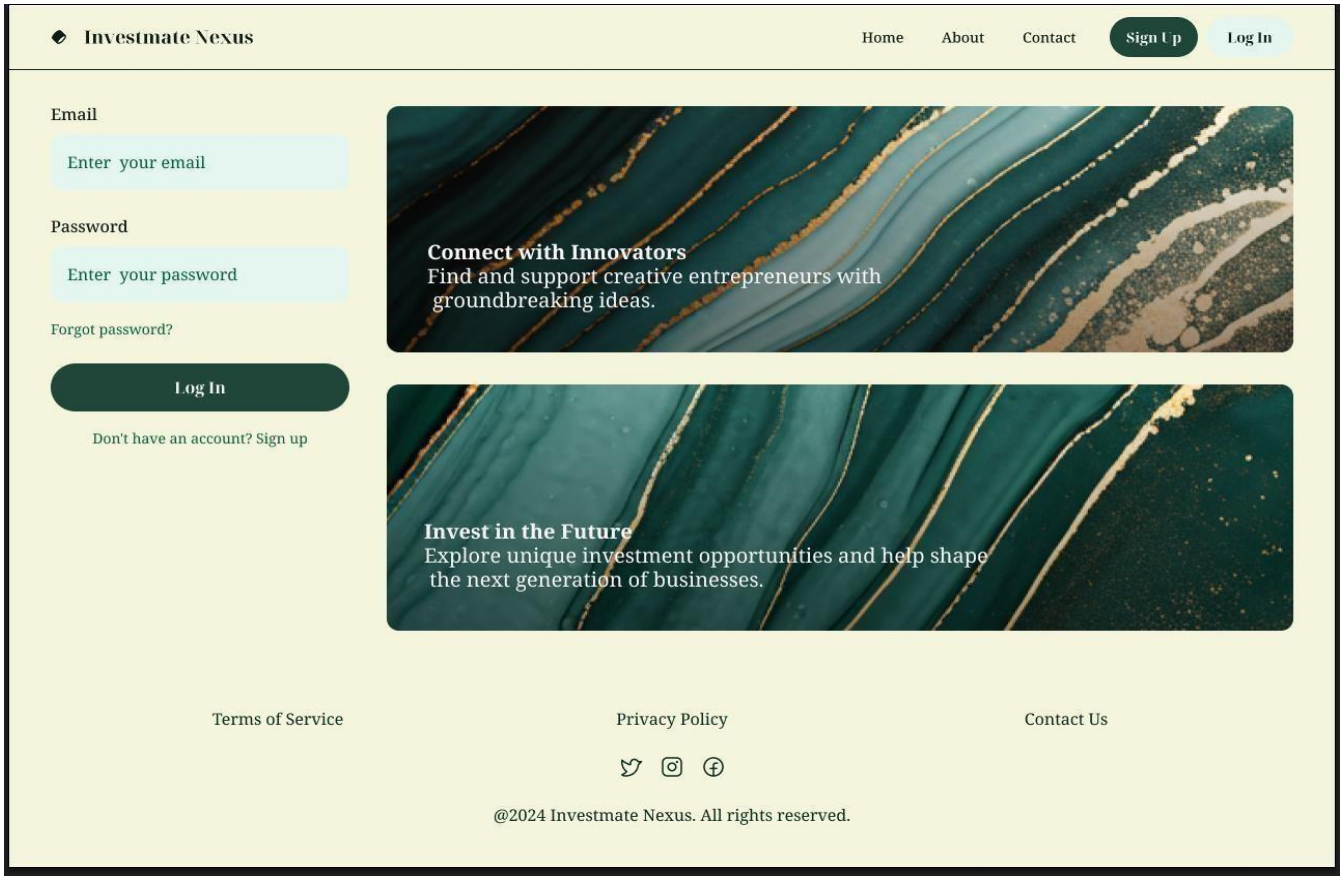
4. Screenshots/Prototype

4.1 Workflow

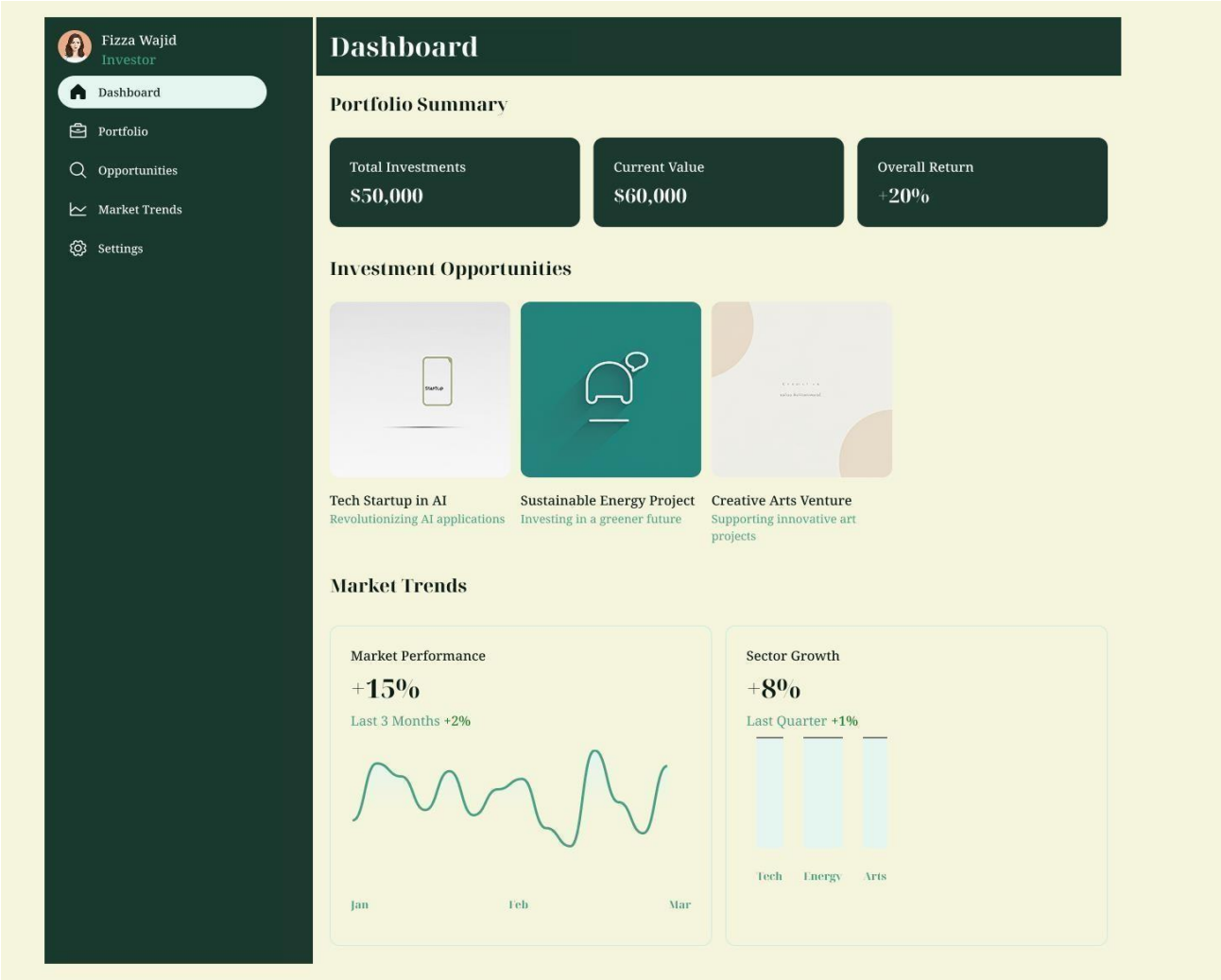


4.2 Screens

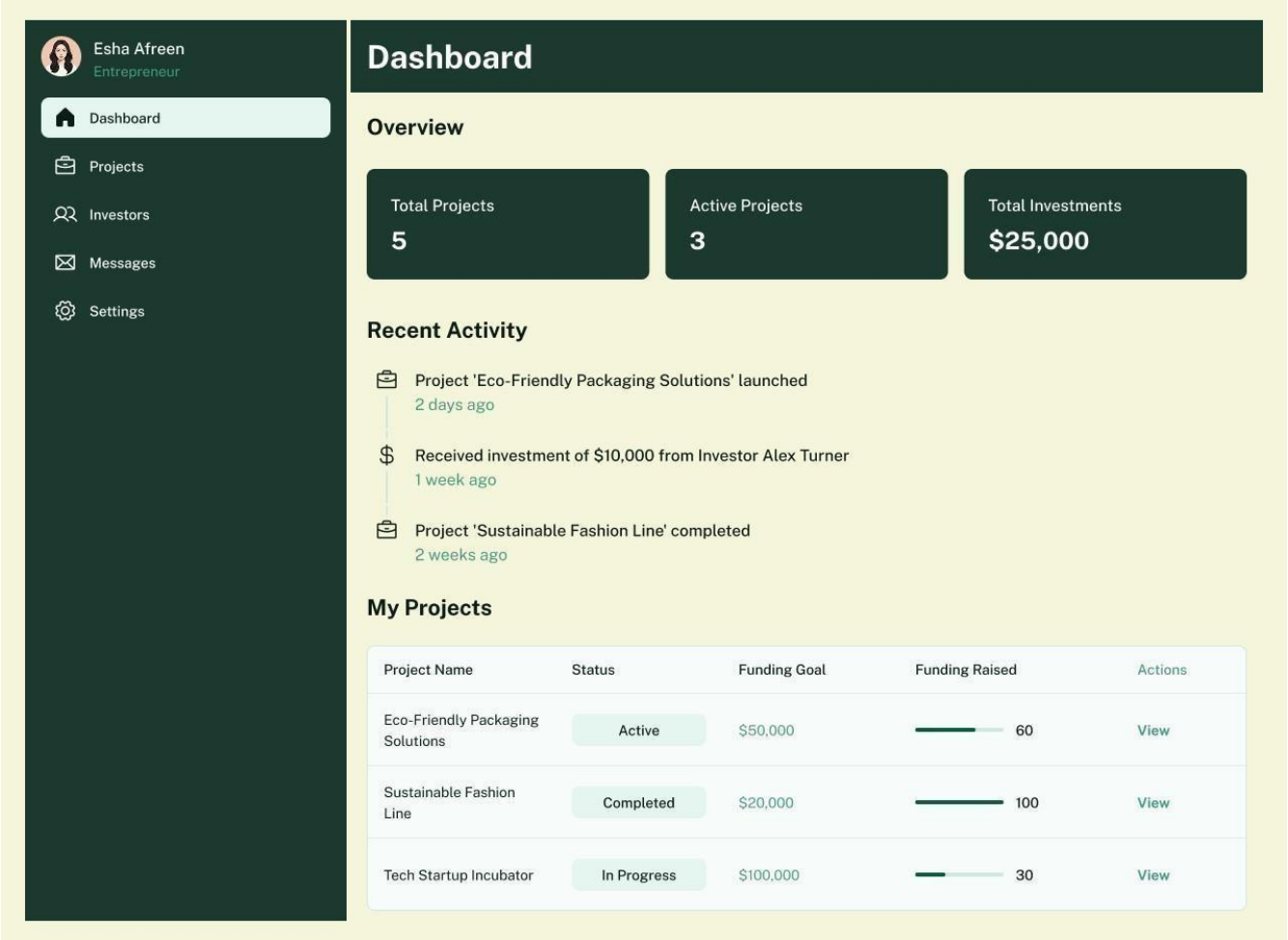
4.2.1. Login/Register:



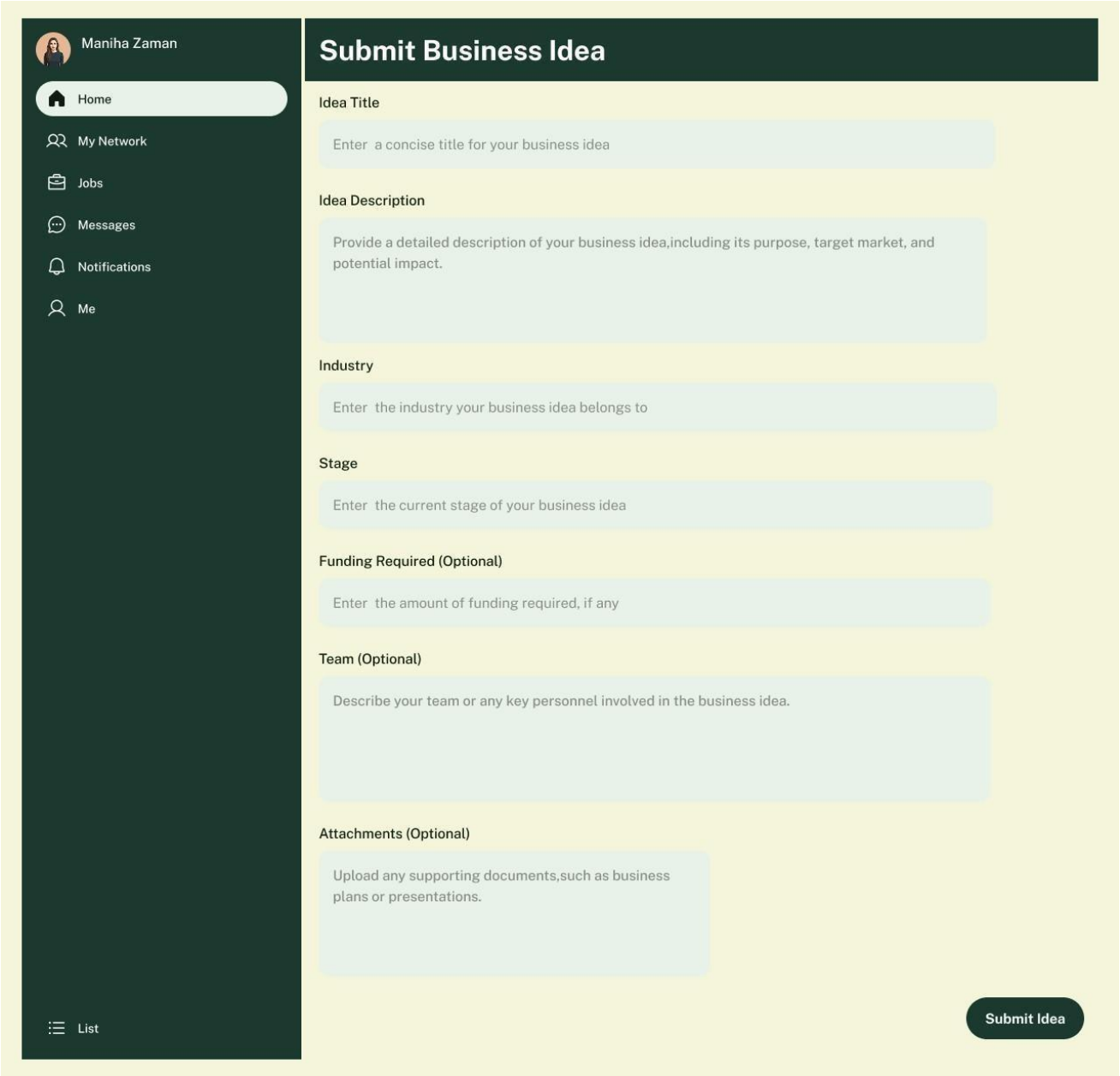
4.2.2. Dashboard For Investor:



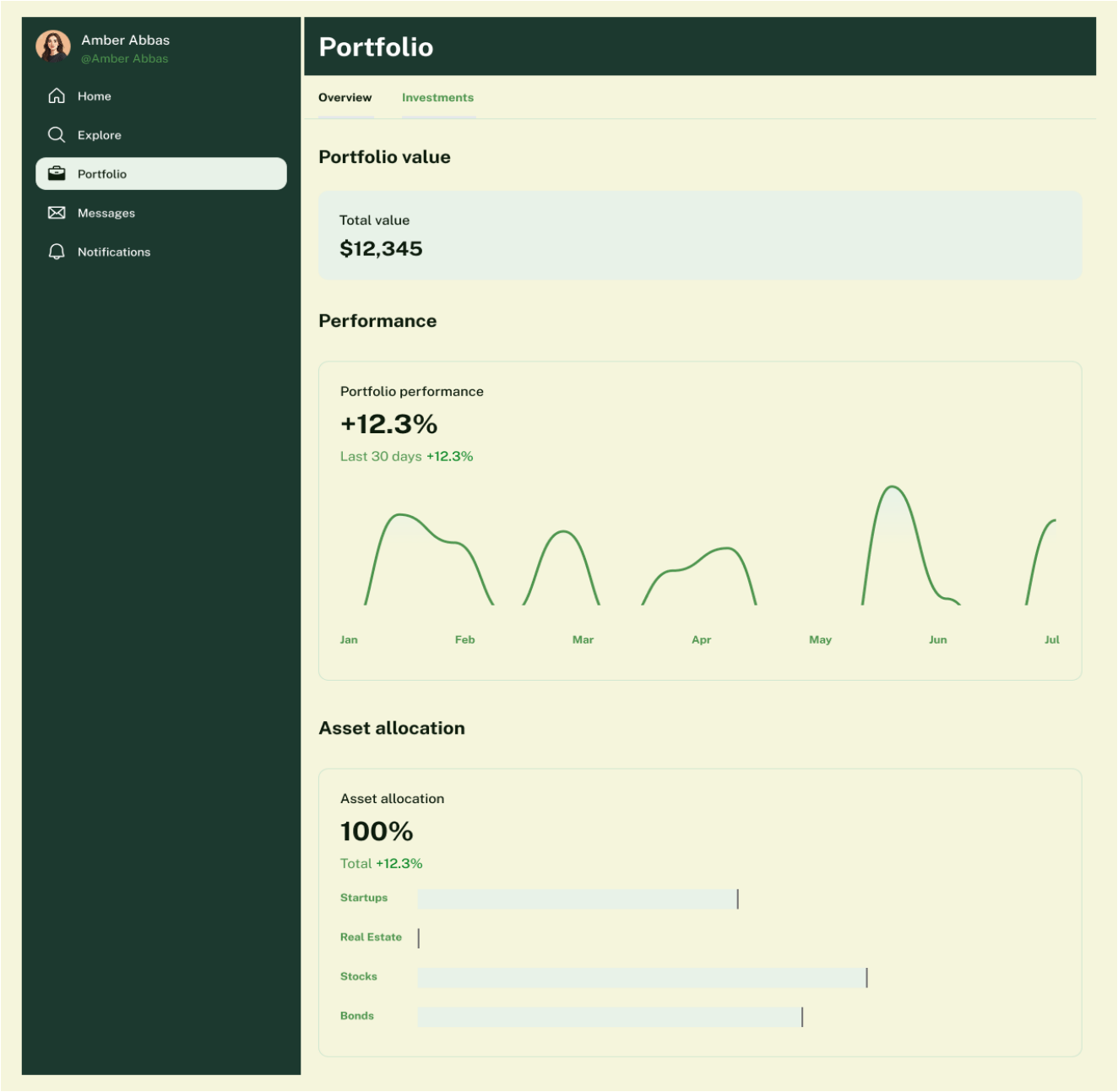
4.2.3. Dashbaord for Entrepreneur:



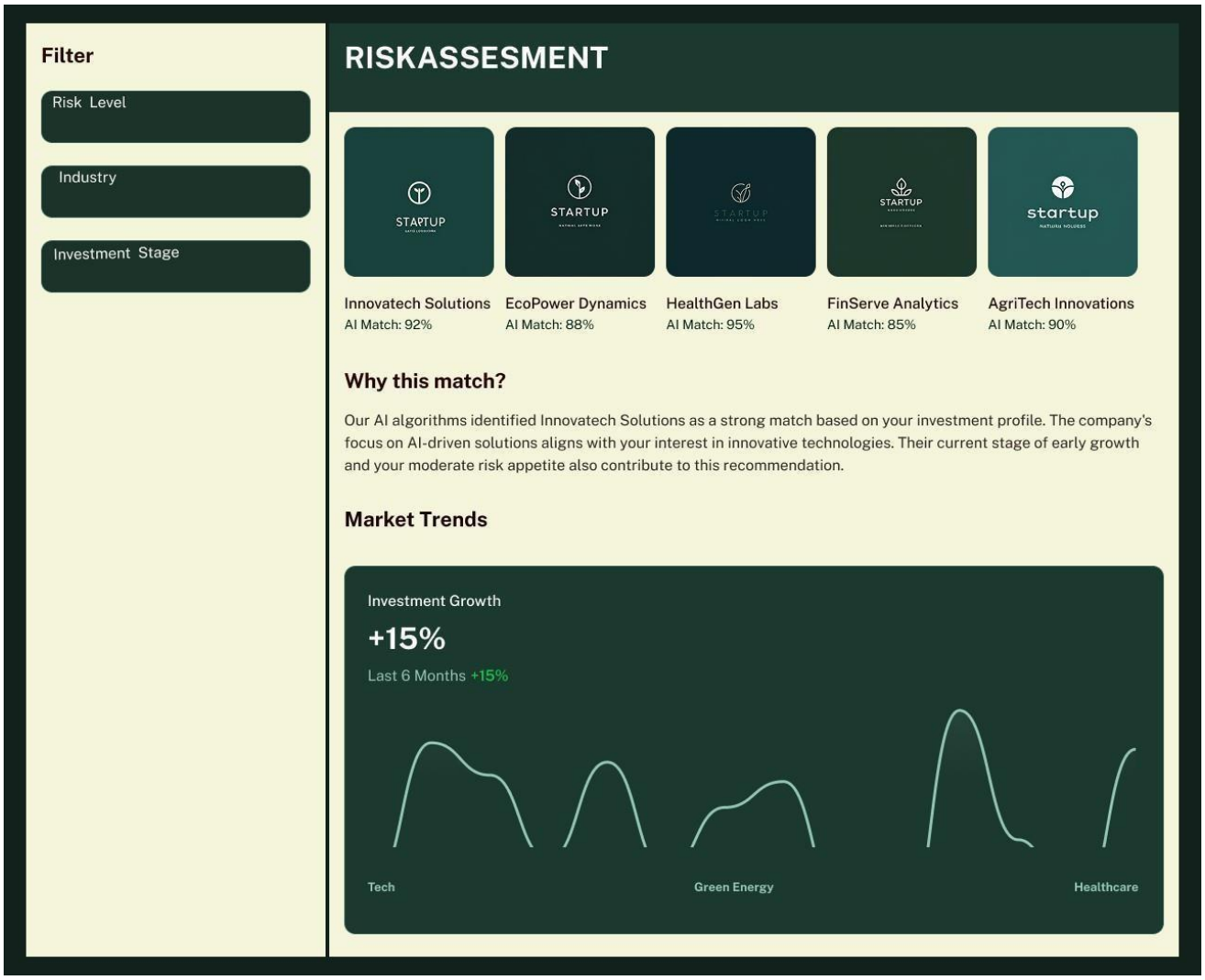
4.2.4. Submit Business Idea:



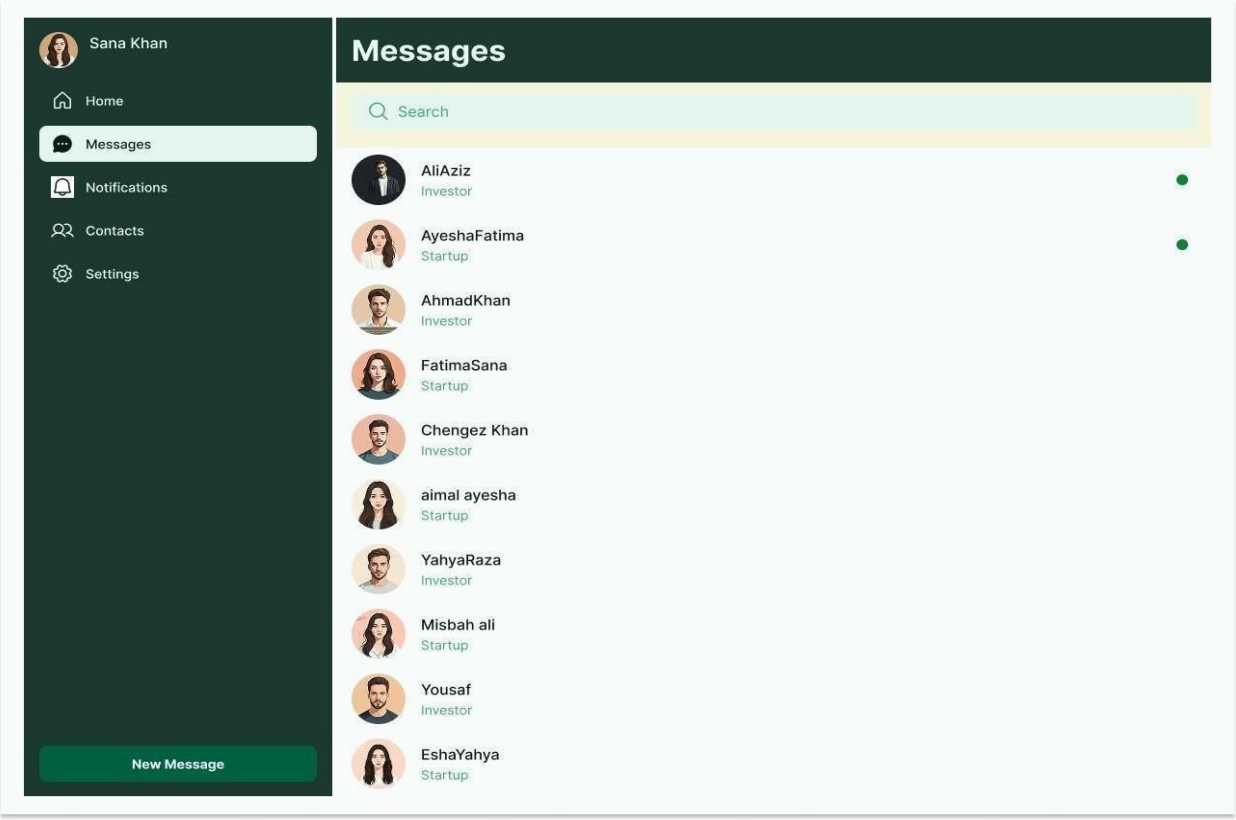
4.2.5. Portfolio Tracker:



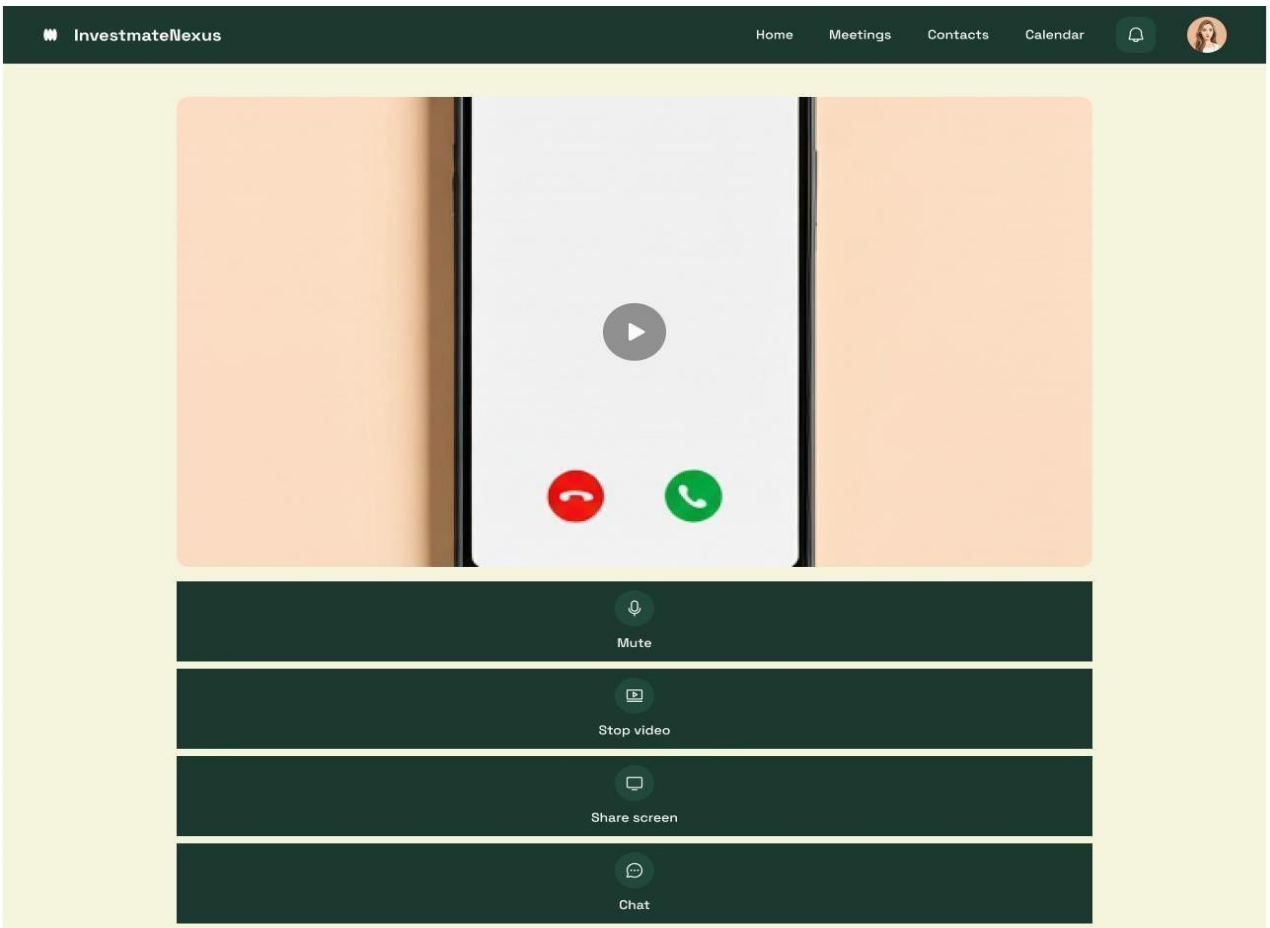
4.2.6. AI recommendations:



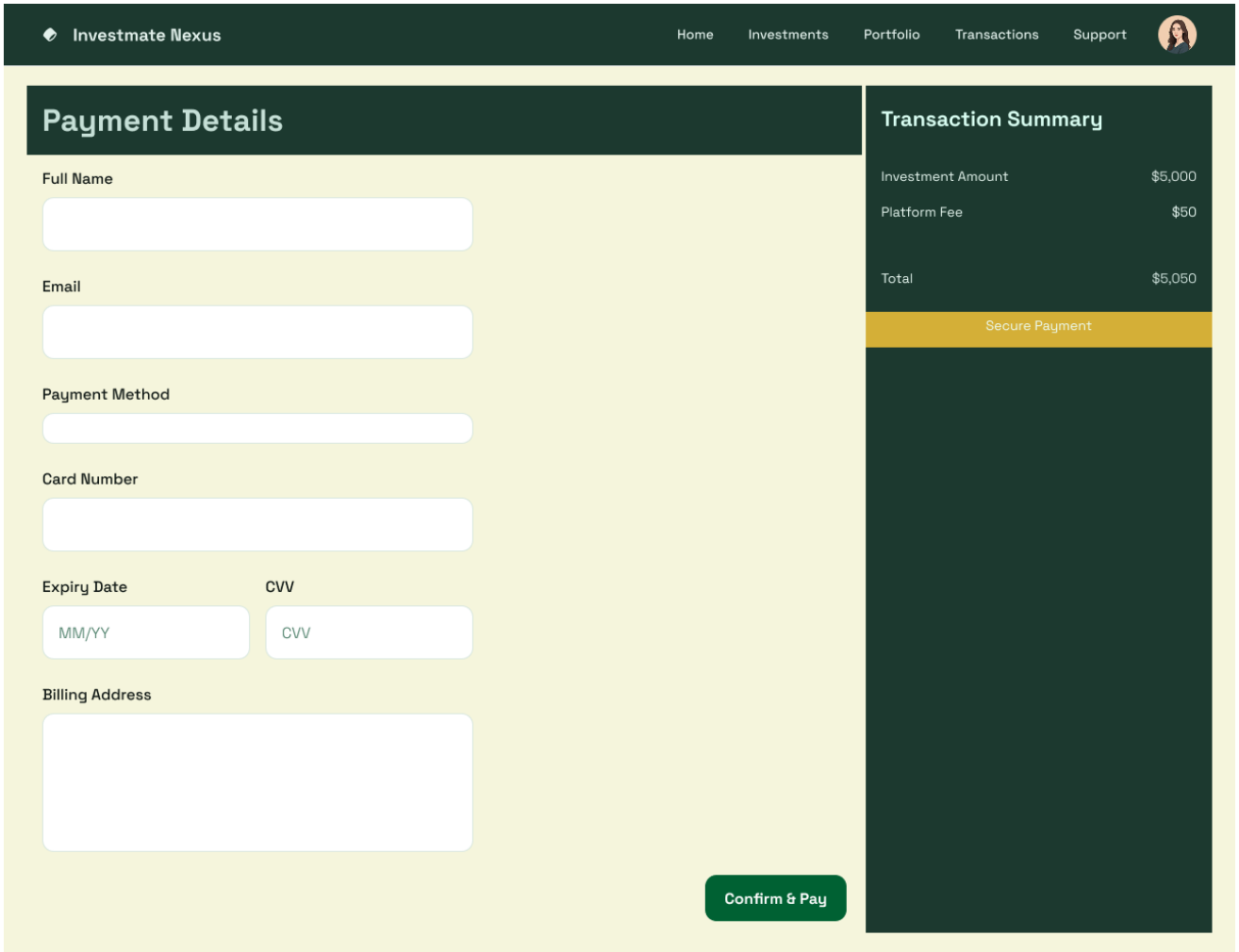
4.2.7. Messaging Interface:



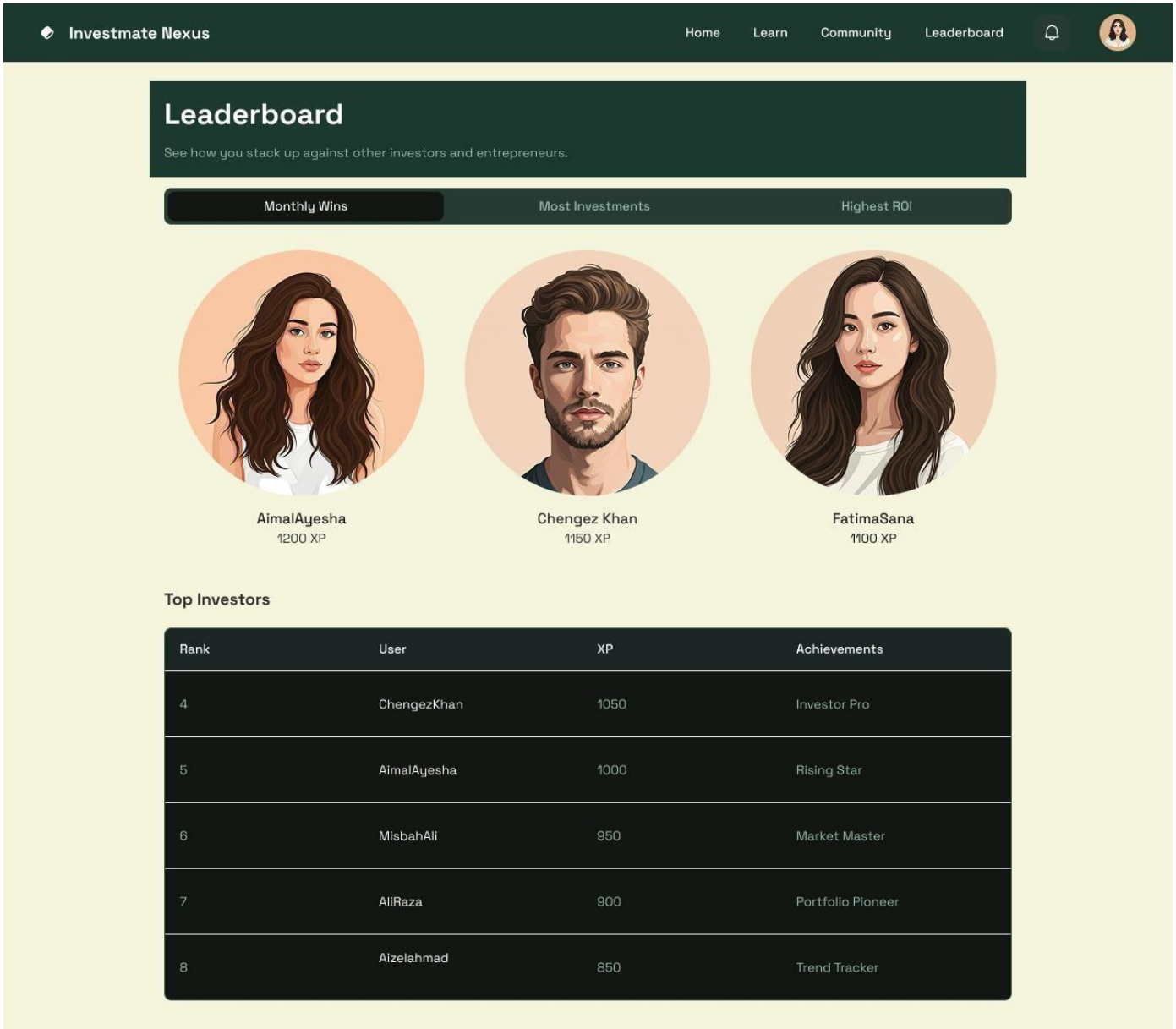
4.2.8. Video Call UI:



4.2.9. Payment Form:



4.2.10. Leaderboard:



4.3 Additional Information

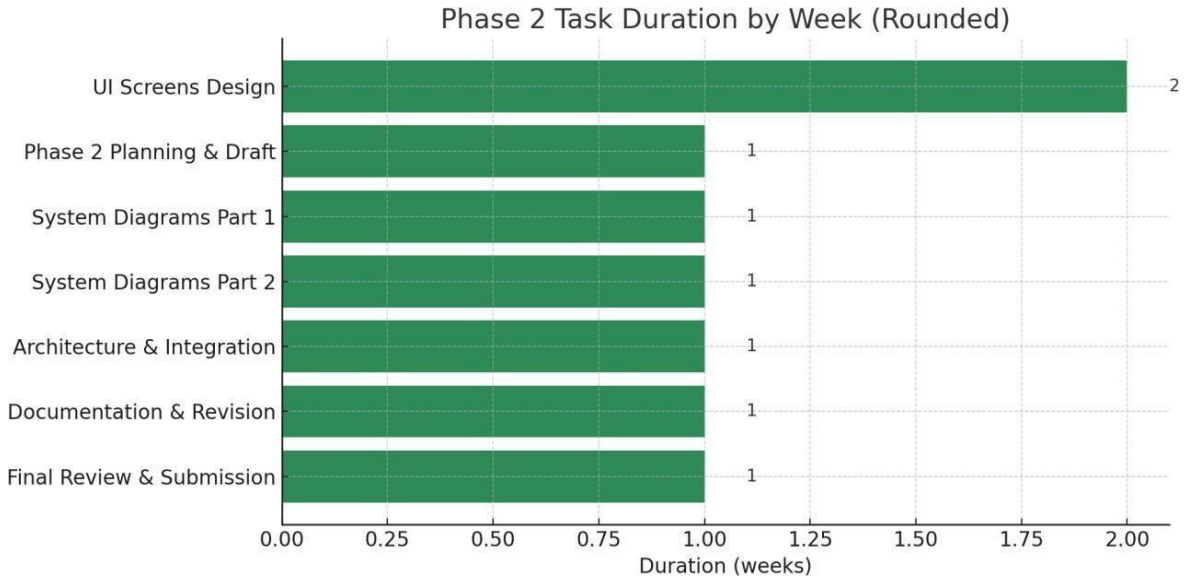
- UI is role-based; each user sees only relevant features.
- Designs are responsive and optimized for mobile and desktop.
- Figma used for screen design and peer-reviewed for clarity.
- Basic form validations (email, password, file types) are included.
- Color scheme (Emerald, Beige, Gold) reflects trust and professionalism.
- Messaging and video call UIs are designed using Twilio mockups.
- Some features (e.g., leaderboard, payment) are partially implemented and planned for future updates.

5. Other Design Details

- **User Interface (UI) Design:**
  - Clean and minimal layout for ease of navigation.
  - Responsive design using Bootstrap to support various screen sizes.
  - Accessible color palette following WCAG 2.1 standards.
- **Technology Stack Used:**
  - Front-End: HTML, CSS, JavaScript, Bootstrap.
  - Back-End (if applicable): Node.js with Express.js.
  - Database (if applicable): MongoDB for flexible, schema-less storage.
- **Version Control and Collaboration:**
  - GitHub used for version control with separate branches for each module.
  - GitHub Actions configured for continuous integration and linting.
  - Pull requests and code reviews ensure quality and avoid conflicts.
- **Testing and Quality Assurance:**
  - Manual and automated testing strategies employed.
  - Browser compatibility testing conducted for Chrome, Firefox, and Edge.
  - Functional validation of all user input and output flows.
- **Challenges and Considerations:**
  - Compatibility issues with certain mobile browsers.
  - Deployment constraints depending on server or hosting provider.
  - Performance optimizations required for larger datasets (if applicable).

6. Revised Project Plan

The project has progressed steadily according to the proposed timeline. Phase 2 involved detailed design and implementation planning, including system architecture, UI design, technical diagrams, and documentation. All tasks were completed on time and reviewed collaboratively by all group members.



7. References

Books:

- "Software Engineering: A Practitioner's Approach," by Roger S. Pressman, 8th Edition, McGraw- Hill, 2014.
- "Database Management Systems," by Ramez Elmasri and Shamkant Navathe, 7th Edition, Pearson, 2015.
- Research Papers and Journals:
- "AI-Powered Recommendation Systems," Journal of Data Science, Vol. 5, Issue 2, 2022.
- "Secure Payment Systems: An Overview," IEEE Research,

2023. Websites:

- AngelList - Connecting Startups with Investors, https://angel.co/, accessed on May 8, 2025.
- Stripe Documentation - https://stripe.com/docs (for secure payment processing).
- Twilio API Documentation - https://www.twilio.com/docs (for messaging and video call integration).
- Payoneer - Global Payment Solutions, https://www.payoneer.com/, accessed on May 7, 2025.
- Real Python - Flask and SQLAlchemy Guide, https://realpython.com/, accessed on May 5, 2025

Appendix A: Glossary

Term	Definition
Entrepreneur	A person who submits business ideas for funding.
Investor	An individual who invests money into business ideas.
Portfolio Management	Feature to track and manage investments.
AI-based Recommendations	Smart suggestions of business ideas for investors.
Risk Assessment	Evaluating risks linked to investments.
Business Idea Validation	Checking the credibility and potential of business ideas.
Secure Payment Integration	Safe and encrypted financial transactions.
Messaging	In-app chat for communication.
Video Call	Real-time online meetings.
Leaderboard	Ranks top investors and entrepreneurs.
Premium Membership	Special access to enhanced platform features.
Payoneer/Bank Transfer	Payment methods for secure transactions.

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 1: List of non-trivial defects

This document has been adapted from the following:

1. Previous project templates at UCP
2. High-level Technical Design, Centers for Medicare & Medicaid Services. (www.cms.gov)